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Functional similarity shapes global patterns of fish naturalization and invasion richness

Zhijun Xia[a], Juanjuan Chen[b,c], Erik Jeppesen[d,e,f], Kun Guo[g], Guohuan Su[h], Guohao Liu[i], Fandong Yu[j,k], Beata Messyasz[l], Jialin Li[b,c], David Z. Zhu[a,m], and Naicheng Wu[b,c,1]

Affiliations are included on p. 11.

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Freshwater ecosystems are among the most threatened by biological invasions, but it remains unclear why some exotics are more likely to be successful (known as naturalization success) and why some communities tend to harbor more exotics than others (measured as invasion richness). Here, using trait-based analysis, we addressed the knowledge gaps based on data from 1,691 fish communities worldwide. We found a latitudinal pattern of naturalization success, showing that successful exotic species were functionally more similar to native species at high latitudes than at low latitudes. Furthermore, the latitudinal gradient of invasion success was driven by contemporary climate, with higher exotic-native functional similarity in stressful climate conditions (i.e., high latitudes), while the opposite was observed in benign climates (i.e., low latitudes). Functional similarity was further positively associated with invasion richness, suggesting that preadaptations to novel environments (i.e., high similarity) may facilitate invasion richness. Moreover, invasion richness increased with increasing basin area and human pressure, but the effect of native diversity on richness was variable. Overall, the results highlight the joint influence of ecological processes (i.e., environmental filtering and niche differentiation) and human activities on fish invasions, with the relative role of ecological processes being well-elucidated through functional similarity.

biological invasion | Darwin's naturalization conundrum | functional traits | riverine fish

Significance

Identifying the factors determining successful invasions is critical for managing exotic species, yet the relative role of ecological similarity, native diversity, and human pressure remains debated. Using global fish community data and trait-based analysis, we simultaneously examine the drivers of naturalization success (naturalization) and invasion richness (proliferation). Our findings reveal a latitudinal gradient: successfully naturalized exotic species are functionally similar to natives at high latitudes but functionally dissimilar at low latitudes. High functional similarity and human pressure tend to facilitate invasion richness, while the strength and direction of native diversity depend on chosen indices. By disentangling the mechanisms underlying naturalization vs. proliferation based on a trait-based approach, this study offers a unified framework moving beyond previous taxonomic studies.

Increasing globalization and human activities have significantly promoted the introduction and naturalization of species beyond their native geographic ranges (1, 2). Such biological invasions, which are surging worldwide, have caused severe ecological and socioeconomic consequences such as biodiversity loss, ecosystem functioning degradation, and substantial economic burdens (3, 4). Freshwater ecosystems are among the most threatened, with over 600 exotic fish species having established self-sustaining populations in more than half of the world's river basins (5). Understanding the key drivers of successful invasion in freshwater ecosystems is thus a major goal in fundamental and applied ecology. However, why some exotics are more likely to be successful (known as naturalization success) and why some communities tend to harbor more exotics than others (measured as invasion richness)–two distinct measures of invasion success–are far from fully understood, especially at the global spatial scale (6, 7).

Traditionally, research has focused on the species-level trait characteristics leading to successful invasions and properties of the native communities that make them more susceptible to invasion (8, 9). Although the results are promising from either perspective, reaching general conclusions has proven difficult. There is now an increasing recognition that instead of considering the two perspectives separately, combining them might yield useful insights, as invasion outcomes may be contingent on ecological similarity or distance between exotic and native species (10, 11). The earliest idea of using ecological similarity as a potential driver of invasion success dates back to Charles Darwin, who proposed two seemingly contradictory hypotheses (12). Initially, Darwin posited that the more closely related an exotic species is to the native community, the more likely it is to be favored during the invasion. This hypothesis is now referred to as Darwin's preadaptation hypothesis, and the rationale behind this is that closely related species tend to share similar functional traits and niche preferences that allow them to preadapt to similar environments (13). Conversely, in his naturalization hypothesis, Darwin proposed that exotic species distantly related to native species tend to be more successful because they would avoid competition and predation and better utilize distinct ecological niches (14). Collectively, the paradox raised by the two opposing hypotheses, emphasizing the dominant role of environmental filtering and niche differentiation, is referred to as Darwin's naturalization

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[1]To whom correspondence may be addressed. Email: naichengwu88@gmail.com.

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conundrum (15, 16). With the increasing availability of phylogeny, Darwin's conundrum has received renewed attention in the past two decades, with phylogenetic similarity being frequently employed to predict invasion outcomes (10, 17). A previous study has revealed that global exotic fish naturalization is enhanced with the presence of closely related native species (5). Nevertheless, phylogenetic similarity may not always be a reliable proxy for functional similarity, as functional and phylogenetic similarity capture distinct ecological and biological processes (e.g., convergent evolution) (18, 19). In this sense, functional similarity appears to be a more promising metric in revealing the mechanisms underlying successful naturalization of freshwater fish, which surprisingly has not been explored at the global scale.

Numerous studies have tested the two opposing hypotheses, yet with conflicting results, and–so far–a clear consensus has not been reached (20, 21). Ecologists have now recognized that the two ecological processes determining invasion success need not be mutually exclusive and that Darwin's conundrum can be reconciled from various perspectives (17, 22, 23). For instance, it is assumed that niche differentiation tends to be more relevant at local spatial scales, while environmental filtering may overwhelm the influence of niche differentiation at large scales (23). However, previous studies conducted at the regional scale have also found that naturalized species tend to be distantly related to the native community (20, 22). Together, these findings suggest that the relative importance of environmental filtering and niche differentiation in determining naturalization outcomes may vary considerably, even at large spatial scales. Similarly, the stress-dominance hypothesis states the spatial variation of the two mechanisms along environmental gradients, with the environmental filtering being more influential under harsh environmental conditions, whereas niche differentiation dominates in benign environments (24). Although this hypothesis is intuitively sound and has received widespread support in community assembly research (25–27), it has rarely been assessed in the context of biological invasion. Prior studies investigating the determinants of plant naturalization success have found that successful exotic species tend to be more closely related to the natives in communities characterized by low temperature, potentially implying a shift of ecological mechanism from niche differentiation to environmental filtering with increasing thermal stress (23, 28). However, previous studies on fish naturalization have generally adopted species-level analysis (5, 6), which cannot readily detect the shift of underlying mechanisms at the community level. Moreover, the nonstandardized distance-based metrics between exotic and native species may not simultaneously distinguish between the two mechanisms of Darwin's naturalization conundrum (5, 7). Additionally, the important role of environmental variables in mediating the relative importance of different processes has been largely ignored in previous studies (5). Thus, whether the two mechanisms underlying naturalization success vary with climate and environmental variation remains unknown for freshwater fish. More importantly, so far, compared with naturalization success, the role of functional similarity in shaping fish invasion richness is still incomplete (7, 29).

Besides exotic-native ecological similarity, multiple abiotic and biotic attributes of the native community may affect the outcomes of biological invasion. For instance, the human activity hypothesis posits that anthropogenic activities promote the invasion success of exotic species by increasing propagule pressure and disturbing natural habitats (30, 31). Moreover, with increased propagule pressure of exotic species exerted by humans, the influence of ecological processes (i.e., niche differentiation and environmental filtering) would be dampened and even masked (32). Additionally, the diversity of native communities has generally been invoked to explain invasion richness. On the one hand, species-rich communities should host fewer exotic species because native species have already filled many empty ecological niches and thus competitively exclude exotic species (i.e., biotic resistance hypothesis) (33, 34). On the other hand, as the environment suitable for native species might also favor exotic species, there should be a positive association between native and exotic species richness (i.e., biotic acceptance hypothesis) (35, 36). Studies examining these hypotheses often use species richness as an indicator of niche filling, which may not fully capture the proportion of niches filled in native communities (7). Instead, functional diversity, considering interspecific variation in life history, ecological, and physiological traits, serves as a more informative metric (32, 37). However, we still lack a unified insight into the effects of native functional diversity on fish invasion richness globally. Furthermore, despite recent advances in investigating fish naturalization success and invasion richness separately (5, 29), a global study that examines the relative importance of functional similarity, native diversity, and human pressure on the two metrics simultaneously is still missing, which hampers our comprehensive understanding of the drivers of fish invasion.

By integrating data on fish occurrence from 1,691 river basins worldwide, a functional trait database, environmental and climatic variables, and human pressure, we assessed the relative role of ecological processes and human activities in shaping two distinct measures of invasion success–naturalization success and invasion richness. We first calculated the standardized effect size of mean functional distance between exotic and native species (sesMFD) in each river basin and employed this metric as naturalization success (*SI Appendix*, Fig. S1). We then assessed the overall distributional trend of sesMFD and how sesMFD varied with latitude, climatic variables, and human pressure. Subsequently, we employed sesMFD, human pressure, environmental variables, and calculated native richness and functional diversity indices to determine their relative influence on the invasion richness of fish communities (*SI Appendix*, Fig. S1). First, we expected that sesMFD would decrease with increasing latitude and the harshness of climate variables because stressful climate conditions tend to filter in exotic species that are functionally similar to native species, and vice versa. Second, we hypothesized that invasion richness would demonstrate a negative correlation with sesMFD, as fish communities dominated by the environmental filtering process are expected to comprise more exotic species due to preadaptation to new environments, and the opposite would be true for those structured by niche differentiation. In addition, we predicted that environmental variation and native diversity would increase with invasion richness as greater environmental heterogeneity supports more exotic species at large spatial scales. We also anticipated that native functional diversity would explain more of the variation in invasion richness compared to native species richness. Finally, we predicted that, due to high propagule pressure, human activities would play a critical role in shaping invasion success.

1. Results

1.1. Naturalization Success. Based on the cleaned dataset, 594 exotic fish species have successfully established in 1,691 of the 3,119 river basins globally (54.2%). The five most widely distributed exotic species were *Cyprinus carpio*, *Oncorhynchus mykiss*, *Salmo trutta*, *Carassius auratus*, and *Gambusia holbrooki*, which have all naturalized in more than 250 river basins. The five most invaded river basins were Colorado, Mississippi, Columbia, Calleguas Creek, and Grande in the United States, each hosting more than 40 exotic fishes (Fig. 1). Exotic-native fish sesMFD

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demonstrated considerable variation across river basins globally (Fig. 1). Overall, sesMFD, based on the realm native pool, did not differ significantly from zero (Wilcoxon test: $P = 0.62$), while the remaining five sesMFD indices across the different species pools were on average lower than zero ($P < 0.05$), exhibiting a functional clustering pattern.

According to the cross-validation approach, boosted regression tree (BRT) analyses accounted for a moderate proportion (20.8 to 39.8%) of the deviance in sesMFD across the six species pools. Based on partial dependency plots, the strength and direction of the influence from different predictors on sesMFD were largely congruent across species pools (Fig. 2). Latitude consistently contributed the most (mean value across species pools: 43.1%) to sesMFD, followed by gross domestic product (GDP) (24.2%), basin area (18.7%), and human footprint index (HFI) (14.1%; Fig. 2). Furthermore, sesMFD generally demonstrated a decreasing pattern with a low latitudinal plateau, implying an initial sesMFD increase, followed by a rapid decrease with latitude. sesMFD was negatively correlated with GDP and positively associated with HFI; however, the relationships were nonlinear (Fig. 2). The influence of basin area on sesMFD was relatively complex. In four out of the six cases, it was positively correlated with sesMFD, while in the remaining two, an initial decrease was followed by an increase (Fig. 2). To explore the underlying drivers of the latitudinal pattern of naturalization success, sesMFD was further related to the two synthetic climatic variables. Across the six species pools, we found that sesMFD was generally positively correlated with the temperature-related variable (PC_temperature), but showed negative correlations with the precipitation-related variable (PC_precipitation) (*SI Appendix*, Fig. S2).

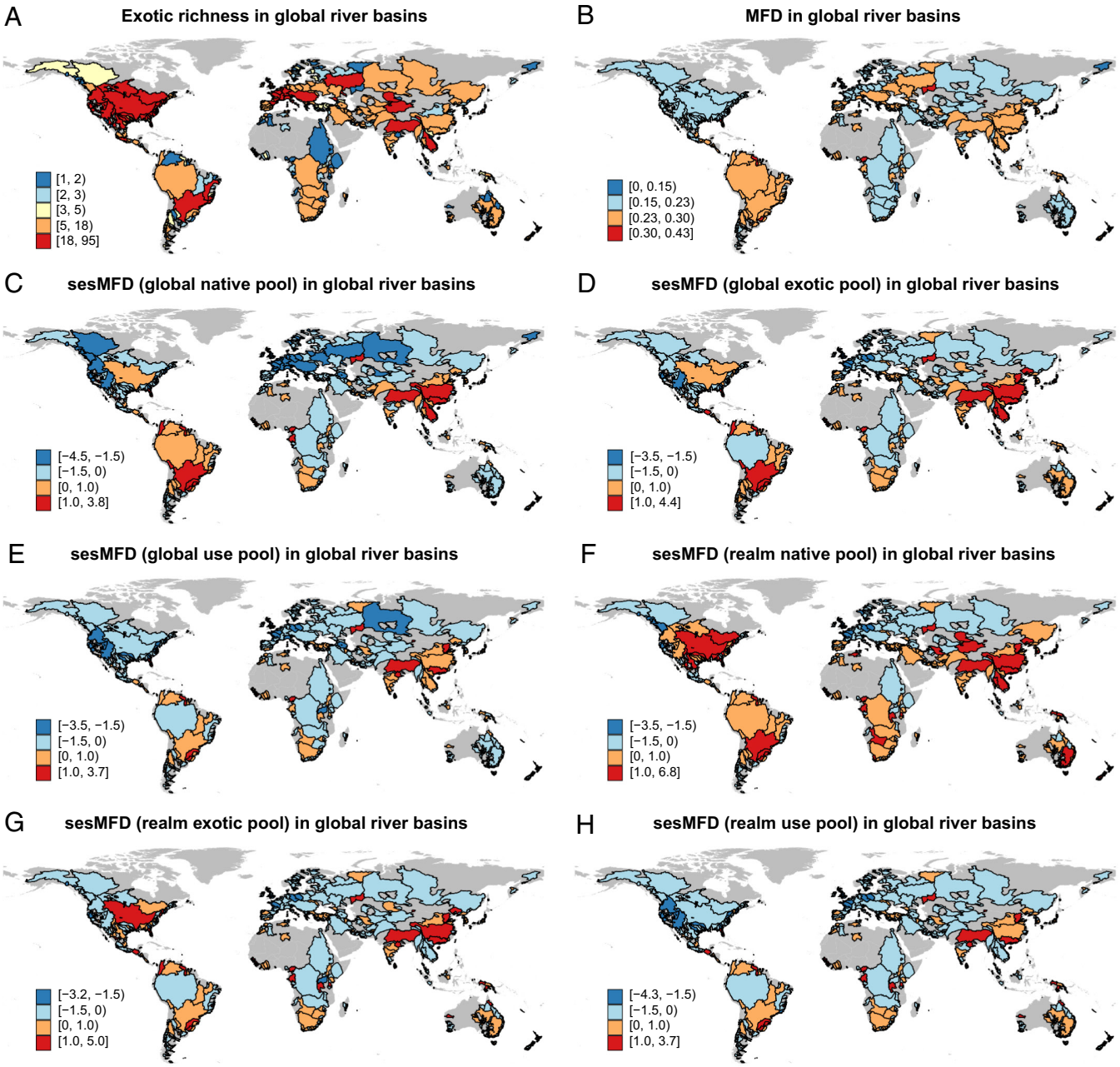


Fig. 1. Global geographic distribution of exotic richness and exotic-native functional similarity of fish species in 1,691 river basins worldwide. (*A*) Geographic variation in exotic fish richness. (*B*) Mean pairwise functional distance (MFD) between exotic and native fish species. (*C*–*H*) The standardized effect size of MFD (sesMFD) is computed based on six different species pools.

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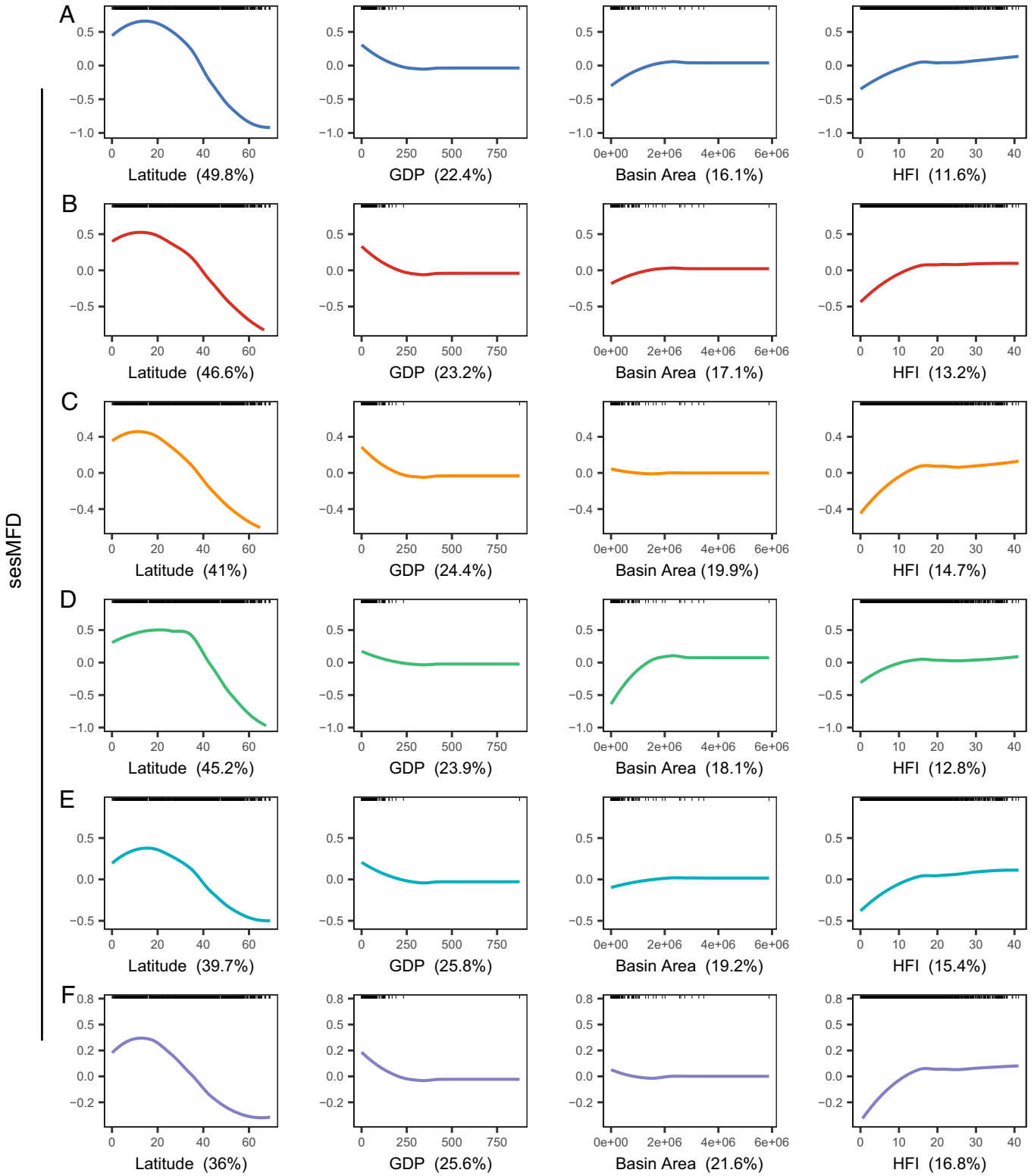


Fig. 2. Results of boosted regression trees showing partial dependency plots between exotic-native functional similarity and predictors related to environmental variables and human pressure. Functional similarity is represented by the standardized effect size of mean pairwise functional distance between exotic and native species (sesMFD) calculated for six different species pools: (*A*) global native pool, (*B*) global exotic pool, (*C*) global use pool, (*D*) realm native pool, (*E*) realm exotic pool, and (*F*) realm use pool. The plots are ranked according to the relative importance of each predictor shown in the parentheses of each panel. Rug plots at the top inside edge of each panel indicate the distribution of river basins along the predictor. GDP: gross domestic product; HFI: human footprint index.

When data were analyzed for each biogeographic realm, the relationship between sesMFD and latitude was generally congruent across species pools, but showed variable patterns in different realms (*SI Appendix*, Fig. S3). Broadly speaking, in Palearctic, Nearctic, Afrotropic, and Neotropic, sesMFD was generally negatively correlated with latitude, consistent with the pattern observed at the global scale (*SI Appendix*, Fig. S3). However, for Indo-Malay and Australasia, a surprisingly positive and U-shaped relationship between sesMFD and latitude was observed, respectively (*SI Appendix*, Fig. S3). Because sesMFD in Indo-Malay and Australasia showed an

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unexpected latitudinal gradient, climatic variables were used to predict sesMFD only in these two realms. In Indo-Malay, sesMFD across species pools often showed a U-shaped relationship with PC_temperature, whereas a hump-shaped or negative association was found for PC_precipitation. These results suggested that climate could not explain the positive latitudinal gradient of sesMFD, and we thus considered two variables (i.e., basin area and elevation range) that are positively correlated with latitude. After fitting the models, sesMFD in Indo-Malay generally increased with basin area and elevation range (*SI Appendix*, Fig. S4). For Australasia, sesMFD across species pools first decreased, then increased, and finally decreased with PC_temperature, but exhibited a positive relationship with PC_precipitation (*SI Appendix*, Fig. S5). These findings indicated that precipitation should be a more important climatic variable than temperature in driving latitudinal variation of sesMFD in Australasia.

1.2. Invasion Richness. Predictors related to functional similarity, environmental variables, human pressure, and native diversity were used to predict invasion richness of fish communities. The BRT analyses explained a comparatively large fraction of deviance in invasion richness globally, with values ranging from 56.0 to 65.1% across species pools. The fitted function between the response variable and each predictor was generally consistent across species pools (Fig. 3 and *SI Appendix*, Fig. S6). As expected, sesMFD was negatively correlated with invasion richness and emerged as the most influential predictor in nearly half of the cases (mean relative contribution across species pools: 21.8%). In the BRT models, basin area was slightly more important than sesMFD (28.1%), demonstrating a positive association with invasion richness. The contribution of latitude to the response variable was moderate (10.0%), with a hump-shaped relationship being found (Fig. 3). Invasion richness increased nonlinearly with GDP, which always ranked as one of the top three important predictors (21.6%). Similarly, HFI was positively correlated with invasion richness, with a mean relative contribution of 5.2% (Fig. 3 and *SI Appendix*, Fig. S6). Although the individual contribution of native diversity indices was low (from 2.8 to 6.5%), their combined influence was appreciable (13.4%). Functional diversity indices explained more variation (native MFR: 5.3%; native MNFR: 4.3%) in invasion richness than did native richness (3.7%). Furthermore, we found native richness and native MFR were positively correlated with invasion richness, whereas native MNFR generally exhibited a negative association with invasion richness (Fig. 3 and *SI Appendix*, Fig. S6). Additionally, interactions between the eight predictors might influence the performance of BRTs, and our results showed that the highest values of invasion richness would be found in river basins with low sesMFD, large basin area, high GDP, and moderate to high latitudes (*SI Appendix*, Fig. S7). Finally, when data were analyzed at the individual realm separately, invasion richness was generally negatively correlated with sesMFD in Palearctic, Nearctic, Afrotropic, and Neotropic, yet a positive pattern was observed in Indo-Malay and Australasia (*SI Appendix*, Fig. S8).

1.3. Sensitivity Analysis. We first tested model results for sensitivity to the inclusion of marine fish species in the occurrence dataset. Functional similarity and invasion richness quantified based solely on freshwater species had similar distribution patterns as those based on all fish species (*SI Appendix*, Fig. S9), with Pearson's correlations between functional similarity higher than 0.97 (*SI Appendix*, Fig. S10). By examining the latitudinal gradient of sesMFD, our sensitivity analysis revealed a low plateau pattern with latitude for sesMFD across species pools (*SI Appendix*, Fig. S11), consistent with those based on all fish species. Furthermore, we found sesMFD was generally negatively related to invasion richness globally, a finding qualitatively similar to that based on all fish species (*SI Appendix*, Fig. S12).

Geographic patterns of invasion richness and functional similarity of fish species with at least one (*SI Appendix*, Fig. S13), five (*SI Appendix*, Fig. S14), and ten (*SI Appendix*, Fig. S15) measured traits were broadly similar to those based on all fish species. Pearson's correlation analysis also confirmed strong correlations ($r > 0.89$) between sesMFD values based on all species and those based on different numbers of measured traits (i.e., one, five, and ten traits; *SI Appendix*, Fig. S10). For the three trait missing scenarios, the influence of latitude on sesMFD and sesMFD on invasion richness were qualitatively consistent with those reported using all species (*SI Appendix*, Figs. S11 and S12). The only exception was that invasion richness showed a U-shaped relationship with sesMFD calculated using species with 10 measured traits and under the constraint of the realm native pool (*SI Appendix*, Fig. S12).

Functional similarity based on river basins with a smaller basin area was also strongly correlated with that based on the full dataset (*SI Appendix*, Fig. S10). Moreover, the relationships between sesMFD and latitude, as well as the relationships between invasion richness and sesMFD, were highly consistent with those recorded using the full dataset (*SI Appendix*, Figs. S11 and S12).

2. Discussion

Globally, exotic-native functional similarity across fish communities was overall higher than random expectations, reflecting functional clustering. This pattern indicates that successful exotic species tend to share similar functional traits with native species in the community. This agrees with a prior fish study applying a phylogenetic approach (5), and both studies emphasize the crucial role of environmental filtering in shaping fish naturalization success. Our results also add to the growing body of evidence that exotic-native ecological similarity can serve as an important predictor of species invasions (20, 21). However, in contrast to the mixed findings (i.e., support for either environmental filtering or niche differentiation) for plant species (20), the prevalence of environmental filtering in determining fish naturalization may be ascribed to two nonmutually exclusive factors. First, the fish database used assembles occurrence records at a relatively large spatial grain (i.e., river basin) rather than from locally sampled communities, which may amplify the effect of environmental filtering. Second, unlike sedentary plants, fish species (or more generally, animals) possess mobility and cognitive abilities, which allow them to actively track suitable habitat patches (5, 38). Therefore, exotic fish species can better adapt to novel environmental conditions in communities that they invade, pertaining to the environment-driven process (13). This does not mean, however, that niche differentiation is not important in driving fish invasions, only that at the scale of our analysis, the signature of environmental filtering becomes more detectable. Indeed, the relative importance of environmental filtering and niche differentiation at the community level may vary across spatial and temporal gradients.

Exotic-native fish functional distance exhibited a negative correlation with latitude at the global scale, which was consistent across different species pools and different scenarios of functional traits missing. This result indicates that exotic species in low-latitude river basins tend to be functionally distinct from native species (functional overdispersion), aligning with the predictions of Darwin's naturalization hypothesis (14). In contrast, in middle- and high-latitude river basins, successful exotic species appear to share similar functional traits with native species (functional clustering), consistent with Darwin's preadaptation hypothesis (13).

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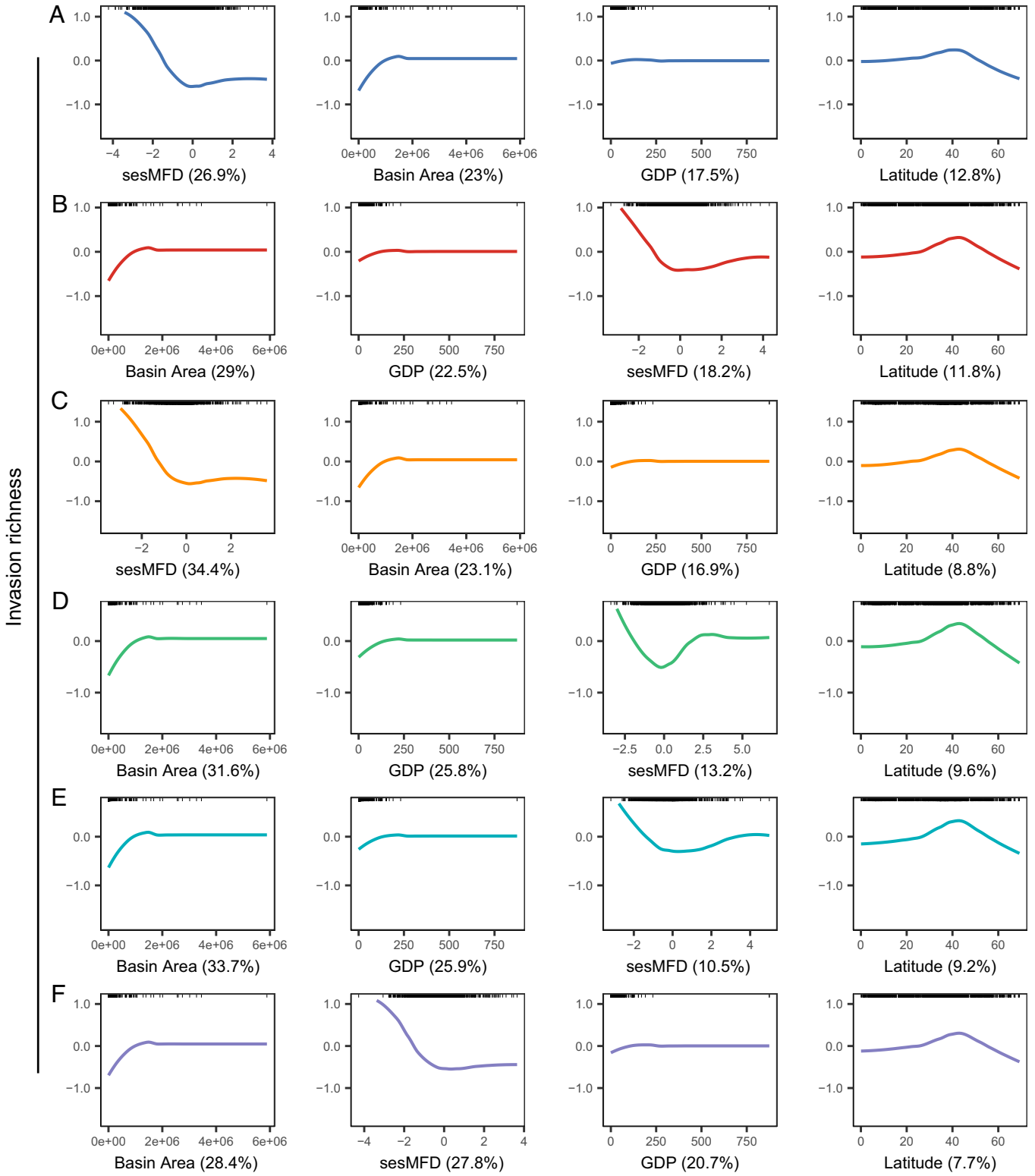


Fig. 3. Results of boosted regression trees showing partial dependency plots between invasion richness and predictors related to exotic-native functional similarity, native diversity, environmental variables, and human pressure. Functional similarity is represented by the standardized effect size of mean pairwise functional distance between exotic and native species (sesMFD) calculated for six different species pools: (*A*) global native pool, (*B*) global exotic pool, (*C*) global use pool, (*D*) realm native pool, (*E*) realm exotic pool, and (*F*) realm use pool. The plots are ranked according to the relative importance of the predictor shown in the parentheses of each panel. Rug plots at the top inside edge of each panel indicate the distribution of river basins along the predictor. GDP: gross domestic product. The figure illustrates the four most important predictors, and refer to *SI Appendix*, Fig. S13 for plots of all predictors.

These results indicate that both sides of Darwin's conundrum might operate simultaneously (15), with their relative importance varying with latitude. Our findings are consistent with a previous work on flowering plant naturalization (28), but extend the insights from the phylogenetic dimension to the functional dimension and to freshwater ecosystems. Using a trait-based approach, this study reports a latitudinal shift of the underlying mechanisms of fish naturalization success. Functional similarity was further related to

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climatic variables, and we found that the latitudinal gradient in Darwin's conundrum for fish was mainly driven by temperature. Specifically, in low-latitude regions (e.g., Southern China, Guyana, and Nepal), the warm and stable climate conditions likely facilitate niche differentiation among native and exotic species, leading to successful naturalized species with various combinations of functional traits. In contrast, high-latitude regions (e.g., Canada and western Europe) have more stressful environments (i.e., cold and large seasonal variation in temperature), which benefits exotic species that are functionally similar to native species, allowing them to establish self-sustaining populations. This finding corroborates previous studies that harsh environmental conditions strengthen the role of environmental filtering in determining species naturalization success (23, 28). Collectively, these results confirm the stress-dominance hypothesis in the context of fish species invasion at the global scale (24).

We found a negative relationship between functional distance and latitude in four of the six biogeographic realms (i.e., Palearctic, Nearctic, Afrotropic, and Neotropic), while a positive and U-shaped latitudinal pattern emerged in Indo-Malay and Australasia, respectively. To disentangle the environmental drivers of the latitudinal gradient, climatic variables were used to explain the variation in functional similarity. For Indo-Malay, we unexpectedly found that both temperature and precipitation variables were not important in driving the latitudinal patterns of functional similarity, despite significant climate-latitude relationships. After searching for other environmental variables that covary positively with latitude, basin area and elevation range stood out as key predictors. The model revealed that functional similarity decreased with basin area and elevation range, suggesting that environmental diversity related to the two variables probably acted as key filters (39, 40). This result may be caused by the comparatively fragmented environments (i.e., existence of many small islands) in this realm. Environmental conditions in small and plain river basins are likely more stressful for fishes than those in large and high-elevation-range rivers due to limited food and habitat resources (19, 31). Following this line of reasoning, successful exotic species need to be functionally similar to the native community in small and plain river basins, consistent with environmental filtering process. Conversely, high environmental diversity in river basins characterized by large areas and high elevation ranges would allow successful invasions by exotic species with various functional strategies. For Australasia, the clear U-shaped latitudinal pattern of functional distance aligns with the U-shaped latitudinal pattern of precipitation. Therefore, precipitation appears to be a more pronounced and important environmental filter determining the naturalization outcomes of fish in Australasia compared to other realms. This inference is further supported by evidence showing that Australasia has the lowest level of precipitation across the six realms (40, 41). Functional distance was positively correlated with precipitation, with communities being clustered and overdispersed functionally in arid (harsh conditions) regions and wet (benign environment) regions. Taken together, the changes in the relative contribution of environmental filtering and niche differentiation along the environmental gradient in each realm also support the stress-dominance hypothesis (24). Our study, therefore, upholds the notion that performing complementary analyses at the realm scale could not only confirm findings from the globe but also reveal exceptions that can provide additional insights into environmental drivers (28).

In addition to naturalization success, we also investigated the patterns and determinants of fish invasion richness globally. Invasion richness was negatively associated with sesMFD calculated from global native pool, global exotic pool, global use pool, and realm use pool, suggesting that river basins with more exotics tended to have exotic and native species that are functionally more similar to each other. However, for sesMFD calculated using realm native and realm exotic pools, invasion richness first decreased but then slightly increased with it. The nonlinear trend could explain why sesMFD based on the realm native pool was not significant in predicting invasion richness in SAR_{error} models. We speculate that this finding results from the increased values of sesMFD when performing the null modeling analyses. Within the same biogeographic realm, exotic species in the null communities probably share similar habitat requirements and functional traits with native species, leading to decreased exotic-native functional distance of random communities and eventually increased sesMFD. Despite this, an overall negative correlation between invasion richness and exotic-native functional distance was observed. Hence, under the strong influence of environmental filtering, functional similarity promotes the accumulation of exotic fish species (7), whereas in communities predominantly structured by niche differentiation, distinct usage of ecological niches tends to constrain exotic fish richness. Thus, we demonstrate that environmental filtering and niche differentiation jointly influence invasion richness through a single metric: ecological similarity, while prior studies, including those for fish species, have typically attributed the observed positive and negative correlations between ecological similarity and invasion success to environmental filtering and niche differentiation, respectively (6, 7, 29). The reason for this is that we used a standardized functional similarity based on a null modeling approach, allowing us to distinguish between the two opposing mechanisms (16), while the nonstandardized metrics used in prior studies can only identify one mechanism at a time. Moreover, the null modeling framework, combined with different definitions of potential species pools, enabled us to control for several well-recognized biases (e.g., failed introduction and introduction bias) in invasion ecology. Thus, the analytical framework used here not only makes our study unique but also has the potential to become a valuable reference for studies of fish invasion at various spatiotemporal scales.

We also examined the relationship between functional distance and invasion richness separately in the six realms and found a negative relationship in Palearctic, Nearctic, Afrotropic, and Neotropic, consistent with the global pattern. Conversely, in Indo-Malay and Australasia, invasion richness increased with functional distance, which suggests that niche differentiation rather than environmental filtering enhances the richness of exotic fishes. These contrasting findings among realms may arise from differences in environmental variables that drive the latitudinal variation in functional similarity. In Palearctic, Nearctic, Afrotropic, and Neotropic, the preadaptation to the thermal (i.e., nonresource) gradient might be more important than niche differentiation in increasing exotic fish richness (28). In Indo-Malay and Australasia, resource availability, however, reverses the role of environmental filtering and niche differentiation in driving invasion richness (42). In arid and small river basins, even if exotic fish species can pass the environmental filter and become naturalized, they have to cope with the scarcity of food and habitat resources, ultimately restricting their proliferation. In contrast, abundant resources in river basins with high precipitation and large areas allow better usage of distinct ecological niches and thus promote the coexistence of multiple exotic fish species, despite niche differentiation. Overall, the findings can be summarized as follows: functional similarity increases invasion richness along a nonresource gradient, supporting Darwin's preadaptation hypothesis, whereas functional dissimilarity promotes invasion richness along a resource gradient, supporting Darwin's naturalization hypothesis. These results offer a perspective to reconcile Darwin's naturalization conundrum in

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predicting invasion richness, which has not been revealed in previous studies (15).

Basin area emerged as one of the most important predictors of fish invasion richness globally, supporting the biotic acceptance hypothesis (35). This suggests that a large basin area, indicative of large habitat size and high habitat heterogeneity (43), tends to promote exotic fish richness. The critical influence of basin area on invasion richness has been verified for fish at both global and continental scales (7, 44). Moreover, we found that native species richness was positively associated with invasion richness, implying that rather than hindering species invasions, species-rich communities are more likely to facilitate exotic richness (45). This observation contrasts with a global study that reported no significant relationship between native and exotic fish richness (31). There might be several reasons for this inconsistency. First, our machine-learning approach can more readily capture the nonlinear relationship compared to linear models (46), potentially amplifying the importance of native richness. Second, our study used a more comprehensive fish database (this study: 13,552 species across 1,691 river basins vs. Leprieur et al.'s study: 9,968 species across 1,055 river basins), and the broader range of exotic richness likely affects the observed pattern (47). Third, we did not include river basins without exotic fishes, while exotic richness in Leprieur et al.'s study represents both invasion presence and pure invasion richness, which might respond differently to native richness (34). Apart from species richness, two functional diversity indices of native communities were also employed to predict invasion richness, albeit with opposite effects. The contrasted influence of native MFR and MNFR on fish invasion may ascribe to their converse relationships with native richness. The former measures functional redundancy across all species, while the latter represents functional redundancy between two similar species. Thus, compared to native MFR, native MNFR should be more affected by the number of species present in the communities. These results indicate that, depending on the chosen native diversity indices, either the biotic acceptance or the biotic resistance hypothesis may be supported, helping to resolve the long-standing invasion paradox (36). Furthermore, functional diversity indices explained more variation in invasion richness than did native richness, which is consistent with our expectations and suggests that functional diversity may more faithfully represent the diversity of functions supported by native communities (7, 48).

Human pressure also markedly influenced fish naturalization success and invasion richness, supporting the human activity hypothesis (30, 31). Specifically, HFI and GDP were positively and negatively correlated with exotic-native functional distance, suggesting that high anthropogenic impacts and economic activities may facilitate naturalization of functionally dissimilar and similar species, respectively (28). These contrasting patterns were probably related to their contrasting associations with latitude. Furthermore, both HFI and GDP showed a positive relationship with fish invasion richness globally, with the effect of the latter being stronger. This contrasts with a recent fish study, which did not include GDP as a predictor of invasion richness (29), suggesting that, compared with other human-related variables, economic activities might be more important for species invasion. The high importance of human pressure in shaping invasion richness may be attributed to three nonexclusive mechanisms: 1) creation of many new niches by disturbing natural habitats, 2) increasing propagule fluxes of exotic species, and 3) increasing the likelihood of unintentional introductions (31). Notably, the influence of human pressure is slightly weaker than functional similarity and basin area, as also demonstrated in a previous study of fish invasion success across US watersheds (7). However, a global-scale study revealed the opposite pattern that GDP exerted a stronger influence on exotic richness compared to basin area (31). Differences in the structure of the used dataset, such as whether or not to exclude rivers without exotic species, may have contributed substantially to the different effects of GDP and basin area on invasion richness. Possibly, GDP plays a more important role in determining the presence of exotic species, while basin area may be a strong correlate of pure exotic richness. Supporting this, a previous study conducted in terrestrial regions (i.e., invasion richness more than zero in each region) rather than river basins found that area had a stronger influence than GDP on fish invasion richness globally (44). Although not directly comparable, we argue that this finding lends support to our inferences. Overall, as the influence of human activities has not often been quantitatively compared to ecological processes, we encourage future studies to adopt an integrative perspective to disentangle their relative contributions to invasion success in diverse ecosystems and taxa.

Several limitations of this study should be acknowledged. First, a functional database with a relatively high proportion of missing traits and a limited number of traits was used. The robustness of our results to trait missing was confirmed by the conducted sensitivity analyses; however, the number and identity of selected traits may still affect the calculated functional diversity and the conclusions (49). Although the ten morphological traits used represent the most comprehensive trait database currently available, they do not cover the entire range of fish functions and cannot fully replace other trait dimensions (e.g., life history and physiological traits) (50). We therefore encourage a further development of functional databases, which could help advance the understanding of the factors determining fish invasion success. Second, frequently, certain exotic fish species are often intentionally introduced in large numbers or with high frequency–referred to as high propagule pressure–increasing their chances of successful naturalization (1, 51). Although we used various human-related variables as proxies for propagule pressure, such interventions might have biased the results (6, 7). To properly address this issue, further studies adopting a multistage framework (i.e., the introduction-naturalization-impact continuum) are needed to better reveal the complex and dynamic processes of biological invasions (37, 52). Third, we employed naturalization success and invasion richness as indicators of invasion success, but the relative importance of ecological and anthropogenic processes may differ when assessing other aspects of invader performance (e.g., abundance and biomass) (32). Moreover, it has been demonstrated that native species performance could be more effective than richness in predicting invasion success (53). Thus, further research integrating multiple aspects of invader and native performance may provide a more holistic understanding of the mechanisms underlying successful invasions. Finally, while our null modeling framework provides robust evidence after controlling for well-recognized biases (e.g., failed introduction, introduction bias, statistical artifact), we acknowledge that temporal dynamic data, including naturalization outcomes and community dynamics, would be ideal for directly testing the drivers of fish invasion success. We thus encourage future studies to compile such data, which will be essential for causal inference.

In conclusion, we conducted a comprehensive evaluation of the effects of functional similarity, native diversity, environmental variables, and human pressure on two distinct measures of invasion success–naturalization success and invasion richness–and found biological invasion was shaped by complex yet complementary mechanisms. We found that successful exotic fishes were functionally more similar to native species at high latitudes than at low latitudes, indicating that with increasing latitude and the

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associated temperature stress, a shift occurred in the underlying mechanisms of naturalization success from niche differentiation to environmental filtering. The opposing effects of functional similarity along the latitudinal gradient provide an insight into solving Darwin's conundrum in predicting naturalization. More importantly, as only a few studies have examined the latitudinal pattern of ecological similarity, there is a pressing need to assess the generality of the findings reported here across other taxonomic groups. In turn, exotic-native functional similarity was positively related to invasion richness globally and in four of six realms, indicating that preadaptation to cold and seasonally variable conditions (i.e., environmental filtering) is more important than niche differentiation for hosting more exotic species. In contrast, along the resource-based environmental gradient in Indo-Malay and Australasia, fish communities structured by niche differentiation were expected to harbor more exotic fishes than those dominated by environmental filtering. Hence, our study offers a perspective to reconcile Darwin's conundrum in predicting invasion richness. Accordingly, our study highlights the critical role of functional similarity in elucidating the ecological processes underlying naturalization success and invasion richness. Additionally, the results reveal that high human pressure and large basin areas facilitate the invasion richness of fish. However, the role of native diversity is somewhat complex and depends on diversity dimensions and the selected indices. Overall, via a trait-based approach, our study advances the current knowledge of mechanisms driving global fish invasions by adding insights across multiple dimensions of invasion success and jointly considering ecological and anthropogenic processes.

3. Materials and Methods

3.1. Global Fish Occurrence Data. We used the currently most comprehensive global database of fish distribution described in (47). The database includes occurrence records of 14,953 fish species in 3,119 river basins, covering more than 80% of the Earth's continental surface. Among the 14,953 fish species, 13,721 species inhabit fresh or brackish waters, while the remaining are marine species entering fresh or brackish waters. Each occurrence is paired with a native or exotic status, with exotic species being defined as successfully introduced species that complete their life cycle and create self-sustaining populations within the river basin (47). Each river basin in the database is given a unique name and is characterized by its location in one of the seven biogeographic realms (i.e., Afrotropic, Australasia, Indo-Malay, Nearctic, Neotropic, Oceania, and Palearctic). This dataset also provides information on geographic coordinates (latitude and longitude) of the river's centroid and river basin area (*SI Appendix*, Fig. S16).

We first standardized the nomenclature of fish species and assigned each species to a family and order according to FishBase (54). Second, for the calculation of functional similarity between exotic and native fish species, we excluded river basins without exotic species (n = 1,400) or native species (n = 3) from the dataset, resulting in 1,716 river basins and 13,718 species. Because the low exotic species richness in Oceania precludes obtaining an adequate number of iterations for the null modeling approach, we also excluded all occurrence records in this realm, resulting in 1,710 river basins. Finally, we removed 19 river basins containing only a single native species due to the inability to estimate native functional diversity. Consequently, our final cleaned occurrence dataset includes 13,552 fish species across 1,691 river basins, encompassing 594 exotic species and 12,958 native species.

3.2. Functional Traits. We compiled functional traits data from the FISHMORPH database, the most comprehensive functional database for fish species worldwide (50). Each species was described morphologically using ten traits that characterize the size and shape of fish body parts, including maximum body length, body elongation, body lateral shape, caudal peduncle throttling, oral gape position, pectoral fin vertical position, pectoral fin size, relative eye size, relative maxillary length, and vertical eye position. Among these, maximum body length, a measure of body size, is a key functional trait related to metabolism and was sourced from FishBase (54). The remaining nine traits describe the shape and position of external anatomical characteristics of fishes related to feeding and locomotion (49) (*SI Appendix*, Fig. S17). These nine traits were computed using 11 morphological measures from side-view pictures of fish and expressed as unitless ratios. The selected functional traits have frequently been applied to evaluate fish functional diversity (55). We acknowledged that these traits cannot capture the full range of fish functions, but they represent the best available trait information on fish species at the global scale (40).

The quality of the fish images did not allow assessment of all 11 morphological measures for all species. Overall, 38.1% of the values were missing in the raw functional database containing 13,552 species. These missing values were evenly distributed among the different taxonomic orders (*SI Appendix*, Fig. S18), although exotic species had a low percentage of missing data (7.8%). To impute missing values, we applied the missForest approach using the missForest package (56) in R version 4.4.2 (57). This machine-learning technique can handle complex interactions and nonlinear relationships between variables and performs better than other algorithms (7, 58). MissForest fills missing values by training a random forest to predict each variable based on the other variables in the dataset. This process is iterative for all variables and stops when the difference between the new imputed and previous imputed matrix becomes large. To improve the accuracy of the imputation, we integrated the first 10 phylogenetic eigenvectors derived from fish evolutionary relationships into the functional dataset before imputation (58). Based on the imputed functional traits data, we log(x + 1)-transformed all traits to improve their normality, followed by scaling (mean = 0, SD = 1) (59). We then conducted a principal component analysis (PCA) and selected the first five PCA axes to compute pairwise Euclidean distance among fish species, as these axes capture a large fraction (97.8%) of the interspecific variation in raw traits. PCA was performed using the stats package in R 4.4.2 (57).

We further tested the phylogenetic signal of functional traits to examine the extent to which traits uniquely reveal the patterns and potential mechanisms of invasion success distinct from phylogenetic relatedness. We calculated phylogenetic signal as Blomberg's K using the phytools package (60). For Blomberg's K, large values (>1) indicate a strong phylogenetic signal. Our results showed that all functional traits had weak phylogenetic signals (*SI Appendix*, Table S1), suggesting that using functional similarity could shed unique insights into studies of fish invasions.

3.3. Exotic-Native Functional Similarity. Based on the calculated pairwise functional distance, we calculated the exotic-native mean pairwise functional distance (MFD) to characterize functional similarity, which quantifies the average functional distance between exotic fish species and all native fishes in a river basin (10, 21). To further examine whether exotic fish species are more or less functionally similar to native species in the resident communities than would be expected by chance, we used a null modeling approach (59). Specifically, we compared the observed exotic-native functional distance to the null distribution of simulated values produced by the null model analysis (28). We generated 1,000 null communities of exotic fish species by randomly shuffling exotic species identity in the potential species pool, while keeping the number of exotic species and native community composition in each river basin constant. This null model has proven more reliable in detecting niche-based assembly processes shaping invasion success than other commonly used null models (16). Thereafter, the standardized effect size (ses) of MFD (sesMFD) was calculated to represent exotic-native fish functional structure, using the following formula:

$$ses = \frac{obs - mean_{null}}{sd_{null}},$$

where obs is the observed value of MFD, and $mean_{null}$ and sd_{null} represent the mean and SD of MFD for 1,000 null communities. A positive ses value indicates that naturalized exotic fish species are functionally more dissimilar to native species than expected by chance, consistent with a functional overdispersion pattern. In contrast, a negative ses value suggests functional clustering and that naturalized exotic fish species are functionally more similar to native species than the null expectation (16).

The definition of the potential species pool may strongly influence the assessment of community functional structure and the interpretation of the observed pattern (61). Therefore, to test the robustness of our results, we considered six different potential species pools of exotic species when calculating the sesMFD

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index (*SI Appendix*, Fig. S19). The potential species pools varied in size, and three were at the global scale and the other three at the realm scale. The first species pool is the largest one and contains all fish species (irrespective of native or exotic status) that are not native to the focal river basin (i.e., global native pool). The second pool considers the possibility that all naturalized exotic fish species in the world have a higher potential to establish self-sustaining populations in the focal river basin (i.e., global exotic pool). Considering fish species in the same realm may have a higher potential to become successfully naturalized (28), we used all native species or exotic fish species of the realm to which the focal river basin belongs as the third (realm native pool) and fourth (realm exotic pool) species pools. Finally, it is generally accepted that humans preferentially introduce fish species with characteristics that provide economic and social benefits (e.g., fisheries, aquaculture, and ornamental importance), and these species are thus more likely to become successful invaders (7, 62). Accordingly, all exotic species with known human use values (sourced from FishBase) at the global scale were included in our fifth species pool, the global use pool. Similarly, our sixth species pool (realm use pool) included exotic species with known human use values within the realm to which the focal river basin belongs.

3.4. Native Community Diversity. Taxonomic diversity of native communities was represented by species richness, which is the number of native fish species within each river basin. We calculated functional diversity metrics encapsulating both richness and relatedness using a tree-based approach (32, 37). First, we built a clustering tree based on the Euclidean distance matrix of the five trait PCA axes by using the Ward clustering algorithm via the hclust function in the stats package (57). We then converted this clustering tree into a functional dendrogram using the ape package (63). For the richness metrics, the functional diversity index provided by Petchey and Gaston was calculated as the sum of branch length of the functional dendrogram for native fishes in each river basin (64). For the relatedness metrics, we calculated mean pairwise functional distance (MFR) and mean nearest functional distance (MNFR) between all native species in each river basin, with a smaller mean distance corresponding to higher redundancy. Native functional richness and relatedness (MFR and MNFR) metrics were assessed through functions in the picante package (65). Because native species richness was highly correlated with the native functional diversity index (Pearson correlation $r = 0.77$), we only kept the former for subsequent analysis, which allows indirect comparison of our results with previous studies. Accordingly, native richness, native MFR, and native MNFR were used to represent native community diversity.

3.5. Climate Variables and Human Activities Indices. To characterize variation in climatic variables along the latitudinal gradient, we used 19 bioclimatic variables from the WorldClim v2.1 database, with a resolution of 1 km^2 (66). These variables measure the mean contemporary climate conditions and the seasonal climatic variability. Due to multicollinearity among the 19 bioclimatic variables, we conducted a PCA to derive synthetic variables. We used the first two PCA axes, accounting for 70.9% of the variation in climatic variables. The first PCA axis (PC_temperature) mainly reflects variation in temperature and its seasonality, while the second PCA axis (PC_precipitation) primarily represents variation in precipitation and its seasonality (*SI Appendix*, Table S2). By running linear models, we found PC_temperature was negatively correlated with latitude ($R^2 = 0.77$, $P < 2e\text{-}16$), whereas PC_precipitation exhibited a quadratic relationship with latitude ($R^2 = 0.14$, $P < 2e\text{-}16$; *SI Appendix*, Figs. S20 and S21).

Given the absence of data on propagule pressure for most fish species (1), human-related variables are usually applied as a proxy (7, 30, 31). Here, we used the human footprint index (HFI), global human modification (gHM) index, and gross domestic product (GDP) as quantitative surrogates for propagule pressure and habitat disturbance (*SI Appendix*, Fig. S21). The HFI integrates data on built environments, croplands, electric infrastructure, navigable waterways, population density, pasture lands, roads, and railways, providing a comprehensive summary of anthropogenic influence on the environment (67). The gHM index assesses the magnitude of impact of human activities (including human settlement, agriculture, transportation, mining, energy production, and electrical infrastructure) on terrestrial systems (68). We also used GDP, which quantifies the size of the economy and is represented by the market value of all final goods and services produced in a region in a given period [in US$, (69)]. The HFI and gHM databases are both available at 1 km^2 resolution; for any grid cell, the former ranges from 0 to 50, while the latter ranges from 0 to 1. GDP values were sourced from (55), having a resolution of 1° of longitude by 1° of latitude. Because HFI was significantly and strongly associated with the gHM index (Pearson correlation $r = 0.86$), we removed gHM from further analysis.

3.6. Statistical Analysis. We applied a one-sample Wilcoxon test to determine whether the overall functional structure (i.e., the six sesMFD metrics across different species pools) differed significantly from zero. All statistical analyses were performed within R version 4.4.2 (57).

To evaluate the relative influence of biotic, abiotic, and human drivers on fish naturalization success (i.e., sesMFD) and invasion richness (i.e., the number of exotic species in each river basin), we applied the boosted regression tree (BRT) analysis (46). BRT does not need prior data transformation or elimination of outliers, can fit complex nonlinear relationships, and can automatically handle interactions between predictor variables. Before formal analysis, we checked for multicollinearity and found no strong correlations between the predictors (*SI Appendix*, Fig. S22).

For the BRT model of naturalization success, we assumed a Gaussian distribution of the response variable and used the methodology proposed by Elith et al. (46). We first included latitude, basin area, HFI, and GDP in the model to examine the variation of functional similarity, as well as the influence of environmental variables and propagule pressure. We further performed BRTs to explore the relative importance of climatic variables (i.e., PC_temperature and PC_precipitation) in shaping functional similarity. In addition, as the geographic pattern of functional similarity may differ across continents, we conducted the above analyses for all six biogeographic realms (28). For the BRT model of invasion richness, we assumed a Poisson distribution of the response variable (7). Specifically, we used eight predictors, including functional similarity, basin area, HFI, GDP, latitude, native richness, native MFR, and native MNFR, to explain variation in invasion richness. We also conducted analyses of invasion richness at the realm scale, similar to those used for modeling naturalization success. For all the above data analyses, we separately fitted BRT models for the six sesMFD computed across six different species pools.

BRTs were fitted using the gbm.step function in the dismo package (70). Analytically, BRT regularization involves jointly optimizing four main parameters: bag fraction (bf), learning rate (lr), tree complexity (tc), and the number of trees (nt) (46). The bf determines the proportion of data drawn randomly from the full training dataset, thereby controlling the stochasticity of BRTs. The lr is employed to shrink the contribution of each fitted tree to the final model, while tc is the number of nodes in a tree that determines the extent to which statistical interactions are fitted. As tc increases, lr must be decreased to fit an adequate number of trees (nt). Here, we selected the optimal model parameters applying a 10-fold cross-validation approach (CV) (46). This offers a parsimonious estimate, CV-D^2 (i.e., the cross-validated proportion of the deviance explained), denoting the expected performance of the optimal model when fitted to new data (46). We explored different combinations of the four parameters and retained the optimal setting of parameters showing the highest CV-D^2. After this, we reran the BRT model 100 times using the optimal parameter setting to overcome potential stochasticity (7). Then, the mean value of the relative contribution of each predictor and the proportion of explained deviance (D^2) of the 100 BRTs were reported. Additionally, interactions among pairwise predictors in BRTs were considered to show the interplay of multiple mechanisms using the gbm.interactions function in the dismo package (70).

Spatial autocorrelation has become an unignorable issue in macroecological and geographical studies over recent decades (71, 72). Because BRTs cannot account for spatial autocorrelation in the response and predictor variables, we also applied autoregressive error (SAR_{error}) models (73). For the SAR_{error} model, all predictor variables were scaled, and quadratic terms were incorporated to account for potential nonlinear relationships. We first extracted the neighbor list based on river basins with continuous boundaries using the poly2nb function and then constructed a spatial weight matrix via the nb2listw function in the spdep package (74). We fitted the SAR_{error} models using the spatialreg package (75), and assessed model performance based on Nagelkerke's R^2 (76). After modeling, we checked for broad spatial autocorrelation in model residuals by calculating Moran's *I* statistic (77). The primary ecological and anthropogenic drivers identified by the SAR_{error} models were generally consistent with BRTs (*SI Appendix*, Tables S3 and S4). The nonsignificant Moran's *I* statistic suggests that spatial autocorrelation did not have a major influence on observed findings.

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We tested the sensitivity of our model results to the inclusion of marine fish species since they likely exhibit different functional trait strategies (47). Therefore, we first removed occurrence records of all marine species and recalculated pairwise functional Euclidean distance among freshwater fish species (n = 12,147). Thereafter, we estimated the standardized effect size of functional distance between exotic and native species for each river basin (n = 1,686) and assessed the correlation between the recalculated functional similarity metric and that based on all species. Finally, we used BRTs to model the relationships between functional similarity with latitude and between invasion richness with functional similarity, two patterns related to our major hypotheses.

Given that a considerable proportion of traits were imputed with a machine-learning algorithm, we conducted three sets of sensitivity analyses to test the robustness of our findings to the missing of traits. We first filtered occurrence records of fish species to include only fish species with at least one (species: n =13,094; river basin: n = 1,690), five (n = 8,104; river basin: n = 1,677), or all ten (n = 6,160; river basin: n = 1,654) measured traits in the functional database. Note that we focused on these three scenarios because the number of species with at least two to nine measured traits was fairly stable (approximately 8,000 species). We then computed the standardized effect size of exotic-native functional distance and assessed whether the results were consistent with those based on all 13,552 fish species. Finally, the effects of latitude on functional similarity metrics and functional similarity on invasion richness were reassessed using BRT analysis.

Finally, we conducted a sensitivity analysis to test the robustness of our results to the river basin scale of the fish occurrence dataset. We removed those very large river basins, resulting in 1,549 river basins (6,512 species) with an average area of 5,872.9 km^2, which is more relevant to examine the related invasion hypotheses (7). Similar to the above sensitivity analyses, we examined the effects of latitude on functional similarity and functional similarity on invasion richness using BRT analysis.

Data, Materials, and Software Availability. All data needed to evaluate the conclusions in the paper are present in the paper. Global fish occurrence data can be retrieved from Figshare (47). FISHMORPH is publicly available in Figshare (50). All other data are included in the manuscript and/or *SI Appendix*.

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Author affiliations: [a]School of Civil Engineering and Future Cities, Ningbo University, Ningbo 315211, China; [b]School of Geography and Remote Sensing, Ningbo University, Ningbo 315211, China; [c]Zhejiang-Germany Joint Laboratory on Remote Sensing of Coastal Ecosystem, Ningbo University, Ningbo 315211, China; [d]Department of Ecoscience and Centre for Water Technology, Aarhus University, Aarhus 8000, Denmark; [e]Sino-Danish Centre for Education and Research (SDC), University of Chinese Academy of Sciences, Beijing 100190, China; [f]Institute for Ecological Research and Pollution Control of Plateau Lakes, School of Ecology and Environmental Science, Yunnan University, Kunming 650228, China; [g]State Key Laboratory of Black Soils Conservation and Utilization, Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences, Changchun 130102, China; [h]State Key Laboratory of Breeding Biotechnology and Sustainable Aquaculture, Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan 430072, China; [i]Department of Geosciences and Geography, University of Helsinki, Helsinki 00014, Finland; [j]Pearl River Fisheries Research Institute, Chinese Academy of Fishery Sciences, Key Laboratory of Prevention and Control for Aquatic Invasive Alien Species, Ministry of Agriculture and Rural Affairs, Guangzhou 510380, China; [k]Key Laboratory of Alien Species and Ecological Security (CAFS), Pearl River Fisheries Research Institute, Chinese Academy of Fisheries Science, Guangzhou 510380, China; [l]Department of Hydrobiology, Institute of Environmental Biology, Faculty of Biology, Adam Mickiewicz University in Poznan, Poznan 61-614, Poland; and [m]Department of Civil and Environmental Engineering, University of Alberta, Edmonton, AB T6G 2H5, Canada

Author contributions: Z.X. and N.W. designed research; Z.X. performed research; Z.X. analyzed data; and Z.X., J.C., E.J., K.G., G.S., G.L., F.Y., B.M., J.L., D.Z.Z., and N.W. wrote the paper.

The authors declare no competing interest.

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